AGRICULTURAL RESEARCH STUDY

Crop Yield Prediction Using Agricultural and Climate Factors in India

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KLE Technological University | Dept of CS&E | 29-04-2026

Introduction

Agriculture plays a vital role in India's economy and supports a large population. Crop yield depends on multiple factors such as rainfall, fertilizer usage, pesticides, soil conditions, and climate.

Due to changing environmental conditions, predicting crop production has become challenging. Machine Learning can analyze historical agricultural data to identify patterns and make accurate predictions. This project focuses on predicting crop yield using various agricultural and climatic features.

Problem Statement

Aerial crop rows

Crop yield varies due to environmental and agricultural factors such as rainfall, fertilizer use, pesticide usage, and cultivation area. Farmers often face uncertainty in estimating production.

> **The goal is to develop a machine learning model that predicts crop yield using historical agricultural data from different states in India.**

Objectives

- To analyze agricultural dataset and understand key features
- To preprocess and clean the data
- To identify factors affecting crop yield
- To build machine learning models for prediction
- To evaluate and compare model performance
- To select the best model for accurate prediction

Dataset Description

Source & Scope

The dataset used in this project is obtained from Kaggle and contains agricultural data from different states in India. It is a structured dataset stored in CSV format.

Scale

The dataset consists of approximately 19,000+ records with 10 features (columns). Each record represents info for a specific crop, year, and region.

Main Features

Crop, Crop Year, Season, State, Area, Production, Annual Rainfall, Fertilizer, Pesticide, and Yield.

Target Variable

Among these, **Yield** is the target variable that the model aims to predict based on environmental and agricultural factors.

Sample Data (From Dataset)

Crop	Year	Season	State	Area	Rainfall	Fertilizer	Pesticide	Yield
Arecanut	1997	Whole Year	Assam	7381	2051.4	702487	2288	0.796
Arhar/Tur	1997	Kharif	Assam	6637	2051.4	631643	2057	0.710
Castor seed	1997	Kharif	Assam	796	2051.4	75755	246	0.238
Coconut	1997	Whole Year	Assam	19656	2051.4	1870662	6119	5238.05
Cotton	1997	Kharif	Assam	1739	2051.4	165500	539	0.420
Dry Chillies	1997	Whole Year	Assam	13587	2051.4	1293075	4211	0.643

Image Sources

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